1 **(a)** Simplify $\frac{2x}{x^0} \div \left(\frac{2y}{x}\right)^{-2}$, leaving your answer in positive indices.

Answer ______________________________ [3]

(b) Express y in terms of p and q, given that $\frac{1}{q} = \frac{2}{y} + p$.

Answer ______________________________ [3]

2 **(a)** Express $\frac{1}{x-3}-\frac{6}{x^2-9}$ as a single fraction in its simplest form.

Answer ______________________________ [2]

(b) **(i)** Express $y = x^2 + 4x - 5$ in the form $y = (x+a)^2 + b$, where a and b are constants.

Answer ______________________________ [2]

(ii) Write down the equation of the line of symmetry of the graph of $y = x^2 + 4x - 5$.

Answer ______________________________ [1]

3 The line AB is shown below.

(a) Construct an equilateral triangle ABC. [1]

(b) Construct the perpendicular bisector of AB. [1]

(c) Construct the angle bisector of $\angle CAB$. [1]

(d) Mark clearly a possible point that is inside the triangle, equidistant from AC and AB, and is nearer to A than B. Label this point P. [1]

(e) The point Q is such that $\angle ACQ = 120°$ and $\dfrac{BQ}{BA} = 0.9$.

Find two possible positions of Q. Label these point Q_1 and Q_2.

Measure and write down the length AQ (where Q can be any of the two points).

Answer ______________________ cm [3]